4.2.8 Vectors

4.2.8.1 Vectors

Content	Additional information

Be familiar with the concept of a vector and the following notations for specifying a vector:
• [2.0, 3.14159, -1.0, 2.718281828]
• 4-vector over $\mathbb{R}$ written as $\mathbb{R}^4$
• function interpretation
  • $0 \mapsto 2.0$
  • $1 \mapsto 3.14159$
  • $2 \mapsto -1.0$
  • $3 \mapsto 2.718281828$
  • $\mapsto$ means maps to
That all the entries must be drawn from the same field, eg $\mathbb{R}$.
	A vector can be represented as a list of numbers, as a function and as a way of representing a geometric point in space.
	A dictionary is a useful way of representing a vector if a vector is viewed as a function.
	$f : S \rightarrow \mathbb{R}$
	the set S = {0,1,2,3} and the co-domain, $\mathbb{R}$, the set of Reals
	For example, in Python the 4-vector example could be represented as a dictionary as follows:
	{0:2.0, 1:3.14159, 2:-1.0, 3:2.718281828}

Dictionary representation of a vector.	See above.
List representation of a vector.	For example, in Python, a 2-vector over $\mathbb{R}$ would be written as [2.0,3.0].
1-D array representation of a vector.	For example in VB.Net, a 4-vector over $\mathbb{R}$ would be written as *Dim example(3) As Single.*
Visualising a vector as an arrow.	For example a 2-vector [2.0, 3.0] over $\mathbb{R}$ can be represented by an arrow with its tail at the origin and its head at (2.0, 3.0).
Vector addition and scalar-vector multiplication.	Know that vector addition achieves translation and scalar-vector multiplication achieves scaling.
Convex combination of two vectors, u and v.	Is an expression of the form $\alpha u + \beta v$ where $\alpha, \beta \geq 0$ and $\alpha + \beta = 1$
Dot or scalar product of two vectors.	The dot product of two vectors, *u* and *v,* $u = [u_1, \ldots., u_n]$ and $v = [v_1, \ldots.., v_n]$ is $u \cdot v = u_1v_1 + u_2v_2 + \ldots\ldots + u_nv_n$
Applications of dot product.	Finding the angle between two vectors.